Ω

$E[X]$
(X)

$f_X(x)$

$f_{X,Y}$
f_X

$E[X|Y]$

∞

P

$\rightarrow$

$1 + 1 = 2$

Ω

Ω		
ω		
A, B		
$P(A)$		
$A \cap B$		
$A \cup B$		

Ω

Ω

$\Omega = \{,\}$

$\{1, 3, 5\}$

$\rightarrow\{HHT, HTH, THH, HHH, \dots\}$

$\rightarrow A^c$

$\rightarrow A \cap B = \emptyset$

$$P(A) \geq 0$$

$$P(\Omega) = 1$$

$$A \cap B = \emptyset, P(A \cup B) = P(A) + P(B)$$

Ω

A

AB

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) \leq P(A) + P(B)$$

$$P(A) =$$

$\rightarrow$

$\rightarrow$

$\times\times$
$10 \times 9 \times 8 = 720$

$3! = 3 \times 2 \times 1 = 6$
$\binom{10}{3} = \frac{720}{6} = 120$

$0.05 0.10$

$$P(A \cup B) \leq P(A) + P(B) = 0.05 + 0.10 = 0.15$$

ΩA

$\times$

$\{2, 3, \ldots, 12\}$
$\{(1,1), \ldots, (6,6)\}$

Ω

BB

$P(A\|B)$		BA
$P(A \cap B)$		AB
		$P()$
		$P(\|)$
		$BAP(A\|B) = P(A)$

$BBB\Omega_{new}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(B) \rightarrow$

$\Omega\{1, 2, 3, 4, 5, 6\}2/6 = 1/3$

$B\{2, 4, 6\}$

$A \cap B\{2\}$

$P(A|B) = \frac{1}{3}$

$$P(A \cap B) = P(B) \cdot P(A|B)$$

BA

$BA_1, A_2, \ldots$

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \ldots$$

BA

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(B)}$$

$$= \frac{\times}{}$$

$P(A) = 0.6P(E|A) = 0.01$

$P(B) = 0.4P(E|B) = 0.02$

$$P(E) = (0.6 \times 0.01) + (0.4 \times 0.02) = 0.006 + 0.008 = 0.014$$

$$P(A|E) = \frac{P(A)P(E|A)}{P(E)} = \frac{0.006}{0.014} = \frac{3}{7} \approx 42.8$$

$P(A \cap B) = P(A)P(B)$

$P(A|B) = P(A)BA$

BA

$ABA \cap B = \emptyset$

AB

$AB \rightarrow$

AB

BA

$P(A)$

$P(B|A)$

$P(A|B)$

$P(A|B)P(B|A)$

$$P(|) \approx 100$$
$$P(|) \approx 0$$

$P(A \cap B) = P(A)P(B)$

$B/P(B)$

Σ

$P(|)$

$P(A|B) = P(A)$

X		$HH \to 2$
$p_X(k)$		k
$F_X(x)$		x
$E[X]$		
(X)		

$\to$

$\Omega \mathbb{R} X(\omega)$

$\Omega = \{HH, HT, TH, TT\}$

X

$$X(HH) = 2$$
$$X(HT) = 1, X(TH) = 1$$
$$X(TT) = 0$$

$\{HH, \ldots\} 0, 1, 2$

$p_X(k)k$

$$\sum p_X(k) = 1$$

$F_X(x)x$

$$F_X(x) = P(X \leq x) = \sum_{k \leq x} p_X(k)$$

$E[X]$

$E[X] = \sum x \cdot p_X(x)$

$$E[X+Y] = E[X] + E[Y]$$

XY

$E[X]$

(X)

$$(X) = E[X^2] - (E[X])^2$$

pnX

$np = 0.1$

$np\times$

pX

$p = 0.01$

$n \to \infty p \to 0$

$\lambda\lambda$

$1(p = 0.01)$

$$E[X] = \frac{1}{p} = \frac{1}{0.01} = 100$$

$\rightarrow$

$n = 50k = 0$

$P(X = 0) = \binom{50}{0}(0.01)^0(0.99)^{50} \approx 0.605$

$\rightarrow$

$\lambda = 5$

$P(X = 0) = e^{-5}\frac{5^0}{0!} = e^{-5} \approx 0.0067$

$\rightarrow$
XY

$(x, y)\sum\sum p_{X,Y}(x, y) = 1$

XY

$E[X + Y] = E[X] + E[Y]$

$\rightarrow$

$\rightarrow$

$\rightarrow$

$\sum\int$

$\sum \int$
$\sum$

Z

	$f_X(x)$	
	$F_X(x)$	$-\infty x$
Z		

$f_X(x)$

y

$$P(a \leq X \leq b) = \int_a^b f_X(x)\, dx$$

$P(X = x) = 0$

$$P(X \geq a) = P(X > a)$$

$$P(T > t + s | T > t) = P(T > s)$$

$e^{-x^2} Z$

$N(0, 1)$

$X \mu \sigma X \sim N(\mu, \sigma^2)$

$$Z = \frac{X - \mu}{\sigma}$$

$Z \Phi(Z)$

$\mu = 50 \sigma = 10$

$z = \frac{70-50}{10} = 2\sigma$

$P(Z \leq 2) \approx 0.977$

$1 - 0.977$

$K \in \{0, 1\}$

X

$X = 0.7 K = 1$

$$P(K = 1 | X = 0.7) = \frac{P(K = 1) \cdot f_{X|K}(0.7|1)}{f_X(0.7)}$$

$\times$

$f_X(0.7)$

$$10 \times 0.1 = 1$$

vs

$\frac{1}{\sqrt{2\pi}\sigma} e^{-\cdots} Z$

xdx

$\rightarrow \rightarrow \rightarrow$

$Z = \frac{X-\mu}{\sigma}$

XY
$\iint$

$XXYXY$

$x, y \int \iint$

$f_{X,Y}(x,y)$		(x,y)
$f_X(x)$		YX
$\iint_A$		A
		(x,y)
	$f_{\Theta\|X}$	

$f_{X,Y}$

x, y

$$P((X,Y) \in A) = \iint_A f_{X,Y}(x,y)\,dx\,dy$$

f_X

YXx

YX

$Y -\infty \sim \infty$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

$X, Y f_{X,Y}(x,y) = f_X(x) f_Y(y)$

$a \le x \le b, c \le y \le d$

$\rightarrow$

$0 \le y \le x \le 1$

$\rightarrow$

XY

$A = \{(x,y) | 0 \le y \le x \le 1\} (X,Y) 1/2 f_{X,Y}(x,y) = 2$

$f_X(x) xyy0x$

$$f_X(x) = \int_0^x 2\, dy = [2y]_0^x = 2x (0 \le x \le 1)$$

$f_X(x) f_Y(y) 2 \ne 2x \cdot 2(1-y)$

$\Theta 0.5$

X

$\Theta \approx 5m$

$5.2m X = 5.2$

$5.2m 5m 5.1m$

$x\Theta$

$$f_{\Theta|X}(\theta|x) = \frac{f_\Theta(\theta) \cdot f_{X|\Theta}(x|\theta)}{f_X(x)}$$

$\theta\times$

θ

$dxdydydx$

ydy

xdx

$f_X(x)xxyy01x$

$\int\int$

Θ

$X, Y Z = X + Y W = X/Y$

$X, Y XY = X^2 XY Z = X + Y$

$X g$

$Y = g(X)$		XY
	$f_X * f_Y$	$X + Y$
ρ		$\sim$
$E[X\|Y]$		Y

$X g(X) Y X$

$F_Y(y)$
$Y X$

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y))$$

$f_Y(y)$
y

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

$y = g(x)$

$$f_Y(y) = f_X(x) \cdot \frac{1}{|g'(x)|}$$

$X g' \to g'$

g'

$1/|g'|$

$Z = X + Y Z$

$X x$

$X + Y = z Y z - x$

$f_X(x) \cdot f_Y(z - x)$

$X x \int$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx$$

$f_Y x \to -x$

z

f_X

$\rho \sim$

$\rho = 1 Y = X$

$\rho = 0$

$\neq$

$\rho = 0$

$X - 1 \sim 1 Y = X^2$

$X Y$

$\rho = 0$

$\rho = 0$

$E[X|Y]$

$E[X]$

$E[X|Y]Y$

$X = Y = E[X|Y]$

$$E[E[X|Y]] = E[X]$$

$E[X]$

Y

$E[X|Y = y]$

$Z = 0.5X + 0.5Y$

$E[Z]$

(Z)

$$(Z) = 0.25(X) + 0.25(Y) + 2(0.5)(0.5)(X, Y)$$

$\rho < 0$

$\int f_X f_Y$

$\rho = 0 Y = X^2$

$E[E[X|Y]]$

t

∞

$P(X \leq 3) n \to \infty$

M_n		$n \frac{X_1 + \cdots + X_n}{n}$
	$\to$	n

$\mu \sigma^2$

$X \geq 0$

$$P(X \geq a) \leq \frac{E[X]}{a}$$

$$P(X \geq) \leq \quad = \frac{1}{10} = 10$$

∞

σ^2

$$P(|X-\mu| \geq c) \leq \frac{(X)}{c^2}$$

cc^2

$nM_n\mu$

$$M_n = \frac{X_1 + \cdots + X_n}{n} \implies (M_n) = \frac{\sigma^2}{n}$$

$n \to \infty \to 0$

μ

S_nN

$S_n = X_1 + \cdots + X_n$

$E[S_n] = n\mu$

$(S_n) = n\sigma^2\sigma\sqrt{n}$

Z

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

$Z_nN(0,1)\Phi(z)$

$X_i \mu \sigma$

$n S_n$

$n = 10000 \mu = 10 \sigma = 5$

$E[S_n] = 10000 \times 10 = 100,000$

$(S_n) = 10000 \times 25 \Longrightarrow \sqrt{250000} = 500$

Z

$$Z = \frac{100,500 - 100,000}{500} = \frac{500}{500} = 1$$

$P(Z > 1) P(Z \leq 1) \approx 0.84$

$$P(S_n > 100500) \approx 1 - 0.84 = 0.16$$

n

$n \geq 30$

$\sqrt{n}$

$S_n n \sqrt{n} n \sqrt{n}$

$\rightarrow$

$\rightarrow$

$\rightarrow$

$\mu \sigma^2$

$\approx$

Z

$n \rightarrow \infty t$

∞

λ		
		p

$t = 1, 2, 3, \ldots p 1 - p$

p

n

n, p

T

p

$$P(T = k) = (1 - p)^{k-1} p$$

$\Delta \to 0$

$pnnp\lambda t$

	p	λ
	$P(K=k) = \binom{n}{k} p^k (1-p)^{n-k}$	$P(k;\tau) = \frac{(\lambda\tau)^k e^{-\lambda\tau}}{k!}$
	$P(T=k) = (1-p)^{k-1} p$	$f_T(t) = \lambda e^{-\lambda t}$

$\lambda_1 \lambda_2$

$$\lambda_{total} = \lambda_1 + \lambda_2$$

$\frac{\lambda_1}{\lambda_1 + \lambda_2}$

λp

λp

$\lambda(1-p)$

$p = 0.01$

$E[T] = \frac{1}{p} = \frac{1}{0.01} = 100$

$\to$

$\lambda = 10$

$P(K \geq 15) = 1 - \sum_{k=0}^{14} \frac{10^k e^{-10}}{k!}$

$\leftrightarrow$

$+ \times p$

π

i		
p_{ij}		ij
P		p_{ij}
π_j		$n \to \infty j$

$X_{n+1} X_n X_{n-1}$

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots) = P(X_{n+1} = j \mid X_n = i) = p_{ij}$$

P

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$$

$n i \to j n P^n$

$p_{ii} = 1$

$n \to \infty$

j

$$\pi_j = \sum_k \pi_k p_{kj}$$

$\pi_j j$

$\sum \pi_k p_{kj} k j$

π

$j \pi_j = \sum \pi_k p_{kj}$

$$\sum_j \pi_j = 1$$

π

$i \mu_i$

$$\mu_i = 1 + \sum_j p_{ij} \mu_j$$

1

$\sum p_{ij} \mu_j j$

a_i

$a_1 = p_{12}a_2 + p_{14}a_4 a_3 = 1, a_4 = 0$

μ_i

$\pi P = \pi$

$x = 1 + \sum px$